HW1

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Q1

Let X, W, and Z denote matrices.

(a)

If $\mathcal{C}(X) = \mathcal{C}(W)$, show that $P_X = P_W$, i.e., the projection matrices must be the same.

Hint: Consider $(P_W - P_X)'(P_W - P_X)$.

(b)

If X is an $n \times n$ matrix of rank n, show $P_X = I_n$ (i.e., the $n \times n$ identity matrix).

(c)

Let $\mathcal{C}^{\perp}(X)$ denote the orthogonal complement of $\mathcal{C}(X)$ (i.e., $\underline{y} \in \mathcal{C}^{\perp}(X)$ if and only if $\underline{x}'\underline{y} = 0$ for any column $\underline{x}$ of X). If a vector $\underline{b} \in \mathcal{C}(X)$ and $\underline{b} \in \mathcal{C}^{\perp}(X)$, show that $\underline{b}$ must equal the zero vector (this also implies that a non-zero vector in $\mathcal{C}(X)$ must be linearly independent from a non-zero vector in $\mathcal{C}^{\perp}(X)$).

Hint: Write $\underline{b} = X\underline{a}$ *and consider* $\underline{b}'\underline{b}$.

(d)

Let X be an $n \times p$ matrix of rank $r \geq 1$, and Z be an $n \times (n - r)$ matrix with columns formed by $n - r$ linearly independent vectors in $\mathcal{C}^{\perp}(X)$. Show that $P_Z = I_n - P_X$.

Hint: Put r *linearly independent columns of* X *along with the columns of* Z *into one* $n \times n$ *matrix* $\tilde{X}$ *(if* $n = r$*, technically* Z *is just a zero vector). Compute* $P_{\tilde{X}}$ *and use (a)-(b).*

(e)

For a vector $\underline{y}$, show that $P_X\underline{y} = 0$ if and only if $\underline{y} \in \mathcal{C}^{\perp}(X)$.

Hint: use (d) and multiply $I = I - P_X + P_X$ *by* $\underline{y}$

(f)

Show that $P_X + P_W$ is a projection matrix if and only if $X'W = 0$ (the columns of X and W must be orthogonal).

Hint: If $P_X + P_W$ *is a projection matrix, one can show* $P_X P_W = -P_W P_X$ *must hold; this relation implies that any vector* $\underline{y}$ *that is in both* $\mathcal{C}(X)$ *and* $\mathcal{C}(W)$ *must be zero (why?). Further, for* $M = P_X P_W$*, the relation also gives* $M'M = P_X P_W P_W$ *and* $M'M = P_W P_X P_W$*, meaning any column of* $M'M$ *belongs to* $\mathcal{C}^{\perp}(X)$ *and to* $\mathcal{C}^{\perp}(W)$.

(g)

Show that $P_X - P_W$ is a projection matrix if and only if $\mathcal{C}(W) \subset \mathcal{C}(X)$.

Hint: If $P_X - P_W$ is a projection matrix then $(I - P_X)P_W = -P_W(I - P_X)$ must hold; then the proof is kind of the same as in (f), using that (from (d)) $P_Z = I_n - P_X$ holds, where Z is a matrix with linearly independent columns that span $\mathcal{C}^{\perp}(X)$.

2.

Suppose that $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and that $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$.

(a)

Show that $AA^{-}C = C$ and $DA^{-}A = D$.

(b)

Using (a), show that

$$\begin{bmatrix} I & 0 \\ -DA^{-} & I \end{bmatrix} \begin{bmatrix} A & C \\ D & B \end{bmatrix} \begin{bmatrix} I & -A^{-}C \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & Q \end{bmatrix},$$

where $Q = B - DA^{-}C$, and I denotes the identity matrix of appropriate dimensions.

(c)

Using (b) with the fact that the two matrices defined by "I on the diagonals" are square and invertible in (b), show that

$$\begin{bmatrix} I & -A^{-}C \\ 0 & I \end{bmatrix} \begin{bmatrix} A^{-} & 0 \\ 0 & Q^{-} \end{bmatrix} \begin{bmatrix} I & 0 \\ -DA^{-} & I \end{bmatrix}$$

is a generalized inverse of

$$M \equiv \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

(d)

Show that the above generalized inverse matches that given in (i) of Theorem 1.4.

(e)

Show that the conditions imposed on the matrices B, C, and D in part (ii) of Theorem 1.4 are satisfied if there exists a matrix $Z = (Z_1, Z_2)$ for which

$$M = Z'Z = (Z_1, Z_2)'(Z_1, Z_2) = \begin{bmatrix} Z_1'Z_1 & Z_1'Z_2 \\ Z_2'Z_1 & Z_2'Z_2 \end{bmatrix} = \begin{bmatrix} A & C \\ D & B \end{bmatrix}.$$

3.

For each of the following statements, indicate whether the statement is true or false, and explain.

(a)

If $Ax = y$ has a unique solution, then A must be a nonsingular matrix.

(b)

If G is a generalized inverse of A, then $GA = I$, where I denotes the identity matrix of appropriate dimensions.

(c)

If A has a generalized inverse G such that $GA = I$, then A must be a (square) nonsingular matrix.

(d)

If A is a symmetric matrix, then all its generalized inverses are also symmetric.

(e)

If A is a symmetric matrix, then it has at least one symmetric generalized inverse.

4.

For each of the following questions, give, if possible, a 3×3 **nondiagonal** matrix A that has the stated property.

(a)

A is a generalized inverse of A.

(b)

A' is a generalized inverse of A.

(c)

The identity matrix I is a generalized inverse of A.

(d)

Every 3×3 matrix is a generalized inverse of A.

(e)

Every generalized inverse of A is nonsingular.

5.

For any matrix A, let B and C be any two matrices that satisfy $A'AB = A'$ and $AA'C' = A$ (the existence of such B and C follows from Corollary 2 to Lemma 1.4). Let $G = CAB$. Show that G satisfies the following four conditions:

(i)

$AGA = A$,

(ii)

$GAG = G$,

(iii)

$(GA)' = GA$, and

(iv)

$(AG)' = AG$.